Shocks, Weak Solutions, and Entropy: An Exposition of Burgers' Equation

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1 Introduction

The viscous Burgers' equation

$$u_t + u\,u_x = \nu\,u_{xx}, \qquad \nu > 0, \tag{1}$$

was introduced by Burgers [2] as a one-dimensional model of turbulence; it had appeared earlier in work of Bateman [1]. It has the form of the Navier–Stokes momentum equation without the pressure term, and combines nonlinear advection with viscous diffusion. Setting $\nu = 0$ gives the inviscid equation

$$u_t + u\,u_x = 0, \tag{2}$$

the scalar conservation law $u_t + f(u)_x = 0$ with flux $f(u) = u^2/2$. Conservation laws of this type arise in gas dynamics and traffic flow [11].

We address the following question: *why does the inviscid Burgers' equation require an entropy condition, and why is that the correct condition?* Section 2 solves (2) by characteristics for C^1 data, and Section 3 computes the time at which classical solutions break down. Section 4 introduces weak solutions and the Rankine–Hugoniot condition, and Section 5 shows that weak solutions are not unique. Sections 6 and 7 introduce the Lax and Oleinik entropy conditions and solve the Riemann problem. Section 8 shows, via the Cole–Hopf transformation, that the entropy solution is the vanishing viscosity limit, and examines Godunov's method numerically. The theory follows [5, §3.4].

2 Method of Characteristics

Consider the initial value problem

$$\begin{cases} u_t + u\,u_x = 0, & x \in \mathbb{R},\ t > 0, \\ u(x,0) = u_0(x), & x \in \mathbb{R}, \end{cases} \tag{3}$$

with $u_0 \in C^1(\mathbb{R})$ in Sections 2 and 3. For discontinuous data, such as the Riemann data (19), (3) is understood in the weak sense of Definition 4.1.

2.1 Characteristics

A *characteristic* of a C^1 solution u is a curve $x(t)$ satisfying

$$\dot{x}(t) = u(x(t), t). \tag{4}$$

Along it,

$$\frac{d}{dt}\,u(x(t),t) = u_x\,\dot{x} + u_t = u_t + u\,u_x = 0. \tag{5}$$

Proposition 2.1 (Characteristics are straight lines)**.** *Let u be a C^1 solution of (3) on $\mathbb{R} \times [0,T)$, and let $x(t)$ solve (4) with $x(0) = x_0$. Then for $0 \le t < T$,*

$$u(x(t),t) = u_0(x_0), \qquad x(t) = x_0 + u_0(x_0)\,t. \tag{6}$$

Proof. By (5), $u(x(t),t) \equiv u_0(x_0)$, so (4) gives $\dot{x}(t) = u_0(x_0)$, which integrates to the stated formula. □

2.2 The implicit solution formula

Theorem 2.2 (Implicit solution via characteristics)**.** *Let $u_0 \in C^1(\mathbb{R})$. For each $t \geq 0$ such that $\Phi_t : \mathbb{R} \to \mathbb{R}$, $\Phi_t(x_0) = x_0 + u_0(x_0)\, t$, is a bijection with C^1 inverse, the function u defined implicitly by*

$$u(x,t) = u_0(x_0), \qquad \text{where } x = x_0 + u_0(x_0)\, t, \tag{7}$$

is the unique C^1 solution of (3) at time t.

Proof. Well-definedness. Bijectivity of Φ_t means $u(x,t) = u_0(\Phi_t^{-1}(x))$ is well defined, and C^1 since Φ_t^{-1} and u_0 are.

The PDE holds. Write $x_0 = x_0(x,t) = \Phi_t^{-1}(x)$. Differentiating $x = x_0 + u_0(x_0)\, t$ in x and in t gives

$$\frac{\partial x_0}{\partial x} = \frac{1}{1 + u_0'(x_0)\, t}, \qquad \frac{\partial x_0}{\partial t} = \frac{-u_0(x_0)}{1 + u_0'(x_0)\, t}, \tag{8}$$

both well defined since $\Phi_t'(x_0) = 1 + u_0'(x_0)t \neq 0$. By the chain rule applied to $u(x,t) = u_0(x_0(x,t))$,

$$u_x = \frac{u_0'(x_0)}{1 + u_0'(x_0)\, t}, \qquad u_t = \frac{-u_0(x_0)\, u_0'(x_0)}{1 + u_0'(x_0)\, t}, \tag{9}$$

so $u_t + u\, u_x = 0$, and $u(x,0) = u_0(x)$ since $\Phi_0 = \mathrm{id}$.

Uniqueness. Let v be any C^1 solution of (3) on $\mathbb{R} \times [0,t]$, and fix $x \in \mathbb{R}$. Solve backwards in time the ODE $\dot{\xi}(s) = v(\xi(s), s)$, $\xi(t) = x$ (the right-hand side is C^1, hence locally Lipschitz, so there is a unique maximal solution). By (5), v is constant along ξ, hence so is $\dot{\xi}$: ξ is a straight line with bounded slope $v(x,t)$. It therefore stays bounded and cannot blow up, so it extends back to $s = 0$. Setting $x_0 = \xi(0)$ we get $v(x,t) = v(x_0,0) = u_0(x_0)$ and $x = x_0 + u_0(x_0)t$. Thus every point (x,t) lies on a characteristic coming from $t = 0$, with $x_0 = \Phi_t^{-1}(x)$, and $v(x,t) = u_0(\Phi_t^{-1}(x)) = u(x,t)$. □

Corollary 2.3 (Local existence)**.** *Let $u_0 \in C^1(\mathbb{R})$ with $m := \inf_{x_0} u_0'(x_0) > -\infty$, and set $T = \infty$ if $m \geq 0$ and $T = -1/m$ if $m < 0$. Then (3) has a unique C^1 solution on $\mathbb{R} \times [0,T)$, given by (7).*

Proof. Fix $t \in [0,T)$ and put $c_t := 1 + \min(m,0)\, t$, so $c_t > 0$ and $\Phi_t'(x_0) = 1 + u_0'(x_0)t \geq c_t$ for every x_0. Three consequences follow. (i) Φ_t is strictly increasing, hence injective. (ii) By the mean value theorem $\Phi_t(x_0) - \Phi_t(0) \geq c_t x_0$ for $x_0 \geq 0$ and $\leq c_t x_0$ for $x_0 \leq 0$, so $\Phi_t(x_0) \to \pm\infty$ as $x_0 \to \pm\infty$; by continuity Φ_t is onto $\mathbb{R}$. (iii) Φ_t' never vanishes, so the inverse function theorem gives a C^1 inverse, depending C^1 on t by the implicit function theorem applied to $x_0 + u_0(x_0)t - x = 0$. Theorem 2.2 therefore applies at every $t < T$. □

Remark 2.4. The uniform lower bound $\Phi_t' \geq c_t > 0$, not strict monotonicity alone, gives surjectivity, and the hypothesis $m > -\infty$ cannot be dropped: for $u_0(x) = x^2$, $\Phi_t(x) = x + tx^2$ is not injective for any $t > 0$, since $\Phi_t'(x) < 0$ for $x < -1/(2t)$.

3 Breakdown of Classical Solutions

3.1 Blow-up of the derivative and the envelope

By the proof of Theorem 2.2,

$$u_x(x,t) = \frac{u_0'(x_0)}{1 + u_0'(x_0)\, t}, \qquad x = x_0 + u_0(x_0)\, t. \tag{10}$$

If $u_0'(x_0) \geq 0$, the denominator is at least 1 and u_x remains bounded along that characteristic for all $t \geq 0$. If $u_0'(x_0) < 0$, the denominator vanishes at $t = -1/u_0'(x_0)$, where $|u_x| \to \infty$ while u remains bounded. Equivalently, the characteristics

$$x(x_0,t) = x_0 + u_0(x_0)\, t, \qquad x_0 \in \mathbb{R}, \tag{11}$$

have an envelope where $\partial x/\partial x_0 = 1 + u_0'(x_0)\, t = 0$, that is, where $\Phi_t'(x_0) = 0$.

3.2 The breakdown time

Proposition 3.1 (Breakdown time)**.** *Let $u_0 \in C^1(\mathbb{R})$ with $m := \min_{x_0} u_0'(x_0) < 0$, attained at $x_0 = a$ (if the minimum is attained at several points, a is any one of them). Then the classical solution exists and is C^1 on $\mathbb{R} \times [0, T^*)$ for*

$$T^* = \frac{-1}{m}, \tag{12}$$

and fails to extend as a single-valued C^1 function beyond T^*. The first breakdown occurs at

$$x^* = a + u_0(a)\, T^*, \qquad t^* = T^*. \tag{13}$$

Proof. Since $s \mapsto -1/s$ is strictly increasing on $(-\infty, 0)$ and $u_0'(x_0) \geq m$ everywhere, $-1/u_0'(x_0) \geq -1/m = T^*$ for every x_0 with $u_0'(x_0) < 0$, with equality exactly at the minimisers of u_0' (in particular at $x_0 = a$). So T^* is the smallest time at which any characteristic meets a neighbour: $\Phi_t'(x_0) > 0$ for all x_0 throughout $[0, T^*)$, and by Corollary 2.3 (with m the infimum of u_0') this gives a classical solution there, while $\Phi_{T^*}'(a) = 0$ makes Φ_{T^*} fail to be locally invertible at a, so u_x is unbounded near (x^*, t^*) by (10). Substituting $x_0 = a$, $t = T^*$ into (11) gives x^*. □

If the infimum of u_0' is not attained, Corollary 2.3 still gives $T^* = -1/m$ as the existence time and $\sup_x |u_x(\cdot, t)| \to \infty$ as $t \uparrow T^*$, by (10) along characteristics with $u_0' \to m$; there is then no first point of degeneracy.

Remark 3.2. At (x^*, t^*) the solution is bounded and continuous but $|u_x|$ is unbounded (a gradient catastrophe); no discontinuity has yet formed.

3.3 Example: $u_0 = -\arctan x$

Here $u_0'(x_0) = -1/(1 + x_0^2) \in [-1, 0)$ is minimised at $a = 0$ with $m = -1$, so Proposition 3.1 gives $T^* = 1$ and $x^* = 0$: the singularity first appears at $(x^*, t^*) = (0, 1)$ (Figure 1). Here $\Phi_t'(x_0) = 1 - t/(1 + x_0^2)$ first vanishes at $t = 1$, at $x_0 = 0$. At $t = 1$ the map $\Phi_1(x_0) = x_0 - \arctan x_0$ is strictly increasing but its inverse is not C^1 at 0; for $t > 1$, $\Phi_t' < 0$ on an interval around $x_0 = 0$, so Φ_t is not injective and characteristics cross.

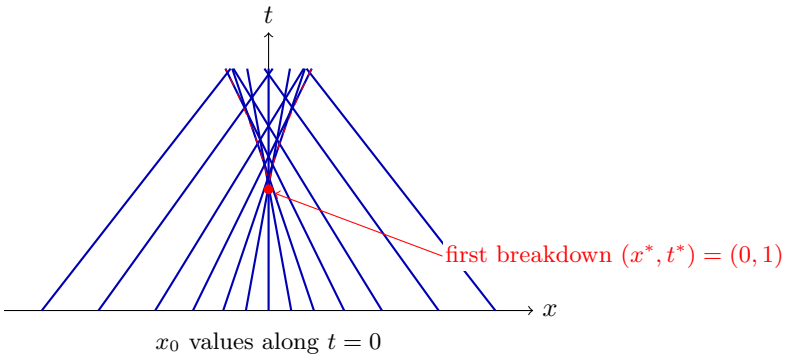


Figure 1: Characteristics $x = x_0 - \arctan(x_0)\, t$ for $u_0 = -\arctan x$, with their envelope (red, dashed), which has a cusp at the first breakdown $(x^*, t^*) = (0, 1)$. Inside the cusp ($t > 1$) three characteristics pass through each point.

Since u_0 is odd and bounded, the entropy solution (which exists by the Lax–Oleinik construction [5, §3.4.2]) is unique (Theorem 6.6), and $v(x, t) = -u(-x, t)$ is an entropy solution with the same data, so the solution is odd in x. Hence for $t > 1$ the discontinuity that replaces the classical solution lies at $x = 0$ and, by (18), is stationary.

4 Weak Solutions

4.1 The integral formulation

Let $f(u) = u^2/2$. If u is a C^1 solution of $u_t + f(u)_x = 0$ with $u(\cdot, 0) = u_0$ and $\varphi \in C_c^1(\mathbb{R} \times [0, \infty))$, then integration by parts in x and t gives

$$0 = \int_0^\infty \int_{\mathbb{R}} \big(u_t + f(u)_x\big)\varphi \, dx \, dt = -\int_0^\infty \int_{\mathbb{R}} \big(u\, \varphi_t + f(u)\, \varphi_x\big) \, dx \, dt - \int_{\mathbb{R}} u_0(x)\, \varphi(x, 0) \, dx. \tag{14}$$

The right-hand side requires no differentiability of u.

Definition 4.1 (Weak solution)**.** Let $u_0 \in L^\infty(\mathbb{R})$. A function $u \in L^\infty(\mathbb{R} \times (0,\infty))$ is a *weak solution* of (3) (with data u_0) if

$$\int_0^\infty \int_{\mathbb{R}} \big(u\,\varphi_t + f(u)\,\varphi_x\big)\,dx\,dt + \int_{\mathbb{R}} u_0(x)\,\varphi(x,0)\,dx = 0 \tag{15}$$

for every $\varphi \in C_c^1(\mathbb{R} \times [0,\infty))$.

Reversing the integration by parts in (14) and applying the fundamental lemma of the calculus of variations shows that a C^1 function is a weak solution if and only if it is a classical solution.

4.2 Piecewise smooth functions and the Rankine–Hugoniot condition

Definition 4.2 (Piecewise C^1 function)**.** Let $s_1, \dots, s_k : [0,\infty) \to \mathbb{R}$ be C^1 with $s_1(t) < \cdots < s_k(t)$ for $t > 0$ (the curves may start at a common point when $t = 0$). Put $s_0 \equiv -\infty$, $s_{k+1} \equiv +\infty$ and $\Omega_i = \{(x,t) :\ t > 0,\ s_i(t) < x < s_{i+1}(t)\}$ for $i = 0, \dots, k$. A bounded function u is *piecewise C^1 with respect to $s_1, \dots, s_k$* if on each Ω_i it is C^1 and u, u_x, u_t extend continuously to $\overline{\Omega_i} \cap \{t > 0\}$. Along $x = s_i(t)$ we write $u_\ell(t)$ and $u_r(t)$ for the limits taken from Ω_{i-1} and Ω_i respectively.

Lemma 4.3 (Boundary identity)**.** *Let u be piecewise C^1 with respect to $s_1, \dots, s_k$ and satisfy $u_t + f(u)_x = 0$ classically in each Ω_i. Then for every $\varphi \in C_c^1(\mathbb{R} \times [0,\infty))$ and every $\varepsilon > 0$,*

$$\iint_{t>\varepsilon} \big(u\varphi_t + f(u)\varphi_x\big)\,dx\,dt = -\int_{\mathbb{R}} u(x,\varepsilon)\,\varphi(x,\varepsilon)\,dx + \sum_{i=1}^{k} \int_\varepsilon^\infty \Big[\big(f(u_\ell) - s_i' u_\ell\big) - \big(f(u_r) - s_i' u_r\big)\Big]\varphi(s_i(t),t)\,dt. \tag{16}$$

Proof. On each $\Omega_i^\varepsilon = \Omega_i \cap \{t > \varepsilon\}$ (a Lipschitz domain, bounded after intersecting with $\operatorname{supp}\varphi$) the vector field $\varphi\,(f(u), u)$ (components in the (x,t) order) is C^1 up to the boundary, and its divergence is $u\varphi_t + f(u)\varphi_x$ because $u_t + f(u)_x = 0$ there. The divergence theorem gives

$$\iint_{\Omega_i^\varepsilon} (u\varphi_t + f(u)\varphi_x)\,dx\,dt = \oint \varphi\,\big(f(u)\,n_x + u\,n_t\big)\,d\sigma,$$

where $d\sigma\,\mathbf{n}$ equals $(1, -s_{i+1}')\,dt$ on the right edge $x = s_{i+1}(t)$, $(-1, s_i')\,dt$ on the left edge $x = s_i(t)$, and $(0,-1)\,dx$ on the bottom edge $t = \varepsilon$. Summing over i, the bottom edges give $-\int u(x,\varepsilon)\varphi(x,\varepsilon)\,dx$, and the two sides of each curve $x = s_i(t)$ (from Ω_{i-1} on its left, Ω_i on its right) combine into the bracket in (16). □

Theorem 4.4 (Rankine–Hugoniot condition)**.** *Let u be piecewise C^1 with respect to $s_1, \dots, s_k$, and a weak solution of (3) for some data $u_0 \in L^\infty$. Then $u_t + f(u)_x = 0$ classically in each Ω_i, and along each curve $x = s_i(t)$, $t > 0$,*

$$f(u_\ell(t)) - f(u_r(t)) = s_i'(t)\,\big(u_\ell(t) - u_r(t)\big). \tag{17}$$

In particular, wherever $u_\ell(t) \neq u_r(t)$, and for $f(u) = u^2/2$,

$$s_i'(t) = \frac{f(u_\ell) - f(u_r)}{u_\ell - u_r} = \frac{u_\ell(t) + u_r(t)}{2}. \tag{18}$$

Proof. Test functions φ supported in a single Ω_i give $\iint (u\varphi_t + f(u)\varphi_x) = 0$, so $u_t + f(u)_x = 0$ in the weak sense there; as u is C^1 on Ω_i, this holds classically, so Lemma 4.3 applies.

Now fix $t_0 > 0$, an index i, and $\varphi \in C_c^1(\mathbb{R} \times (0,\infty))$ supported in a small ball around $(s_i(t_0), t_0)$ meeting no other curve. Take ε below the support of φ in (16): the left side is 0 by (15) (as $\varphi(\cdot,0) = 0$), the bottom term vanishes, and only the curve $x = s_i(t)$ contributes to the sum, so

$$0 = \int_0^\infty \Big[\big(f(u_\ell) - s_i' u_\ell\big) - \big(f(u_r) - s_i' u_r\big)\Big]\varphi(s_i(t),t)\,dt.$$

Since $\varphi(s_i(t),t)$ may be any C^1 function of t with small support, the bracket vanishes identically, which is (17). The first form of (18) follows by division, and the second from $f(u_\ell) - f(u_r) = \frac{1}{2}(u_\ell - u_r)(u_\ell + u_r)$. □

Proposition 4.5 (Piecewise smooth weak solutions)**.** *Let $u_0 \in L^\infty(\mathbb{R})$ and let u be piecewise C^1 with respect to $s_1, \dots, s_k$. Suppose*

(i) *$u_t + f(u)_x = 0$ classically in each Ω_i;*

(ii) *the jump relation* (17) *holds along each curve* $x = s_i(t)$, $t > 0$ *(automatic where* u *is continuous across the curve);*

(iii) $u(x,t) \to u_0(x)$ *as* $t \to 0^+$ *for every* $x \notin \{s_1(0), \dots, s_k(0)\}$.

Then u *is a weak solution of* (3) *with data* u_0.

Proof. Fix $\varphi \in C_c^1(\mathbb{R} \times [0,\infty))$ and $\varepsilon > 0$. By (i), Lemma 4.3 applies, and by (ii) every term of the sum in (16) vanishes, so

$$\iint_{t>\varepsilon} \big(u\varphi_t + f(u)\varphi_x\big)\,dx\,dt = -\int_{\mathbb{R}} u(x,\varepsilon)\,\varphi(x,\varepsilon)\,dx.$$

Let $\varepsilon \to 0^+$. The left side tends to the integral over $t > 0$ (bounded integrand, compact support). On the right, $\varphi(\cdot,\varepsilon) \to \varphi(\cdot,0)$ uniformly and, by (iii), $u(\cdot,\varepsilon) \to u_0$ almost everywhere with $|u| \le \|u\|_\infty$, so dominated convergence gives $-\int u_0\varphi(\cdot,0)\,dx$. This is (15). The curves need not be separated at $t = 0$, since the argument is run on $t > \varepsilon$. □

5 Non-Uniqueness of Weak Solutions

Consider (3) with the Riemann data

$$u_0(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0. \end{cases} \tag{19}$$

Characteristics emanate from $x_0 < 0$ with speed 0 and from $x_0 > 0$ with speed 1, so none covers the wedge $0 < x < t$.

Proposition 5.1. *The function*

$$u_A(x,t) = \begin{cases} 0, & x < t/2, \\ 1, & x > t/2, \end{cases} \tag{20}$$

is a weak solution of (3) *with data* (19).

Proof. Apply Proposition 4.5 with $k = 1$ and $s_1(t) = t/2$: u_A is constant on each side, the jump moves at the Rankine–Hugoniot speed $(0+1)/2 = 1/2$ of (18), and $u_A(\cdot,t) \to u_0$ at every $x \neq 0$. □

Proposition 5.2. *The function*

$$u_B(x,t) = \begin{cases} 0, & x \le 0, \\ x/t, & 0 < x < t, \\ 1, & x \ge t, \end{cases} \tag{21}$$

is a weak solution of (3) *with data* (19).

Proof. Apply Proposition 4.5 with $k = 2$, $s_1 \equiv 0$, $s_2(t) = t$: u_B is constant on the outer regions, solves the PDE on $0 < x < t$ since $(u_B)_t + u_B(u_B)_x = -x/t^2 + (x/t)(1/t) = 0$, is continuous across both curves (so (17) holds trivially), and $u_B(\cdot,t) \to u_0$ at every $x \neq 0$. □

Corollary 5.3 (Non-uniqueness)**.** *The initial value problem* (3) *with data* (19) *has at least two distinct weak solutions,* u_A *and* u_B*, which differ on the wedge* $0 < x < t$.

Remark 5.4 (An infinite family). For every $\alpha \in \mathbb{R}$ let

$$u^\alpha(x,t) = \begin{cases} 0, & x < \frac{\alpha}{2}t, \\ \alpha, & \frac{\alpha}{2}t < x < \frac{1+\alpha}{2}t, \\ 1, & x > \frac{1+\alpha}{2}t. \end{cases}$$

The jumps $0 \to \alpha$ and $\alpha \to 1$ move at their Rankine–Hugoniot speeds $\alpha/2 < (1+\alpha)/2$, so Proposition 4.5 (with $k = 2$) shows that each u^α is a weak solution with data (19). These are distinct for distinct α, except that $u^0 = u^1 = u_A$.

6 The Entropy Condition

6.1 The Lax shock condition

Definition 6.1 (Lax shock condition)**.** Let u be a weak solution of a scalar conservation law $u_t + f(u)_x = 0$ with f convex, and suppose u has a jump discontinuity along a C^1 curve $x = s(t)$ with left and right limits $u_\ell(t)$, $u_r(t)$. The jump satisfies the *Lax shock condition* if

$$f'(u_\ell(t)) > s'(t) > f'(u_r(t)) \tag{22}$$

for all t.

For $f(u) = u^2/2$ and $s' = (u_\ell + u_r)/2$, (22) reduces to

$$u_\ell > \frac{u_\ell + u_r}{2} > u_r \quad \Longleftrightarrow \quad u_\ell > u_r. \tag{23}$$

Thus characteristics on both sides run into the shock. The Lax condition constrains jumps only; a derivation of $u_\ell > u_r$ from viscosity is given in Proposition 8.6.

6.2 Entropy solutions: the Oleinik condition

Definition 6.2 (Entropy solution)**.** A weak solution $u \in L^\infty$ of (3) is an *entropy solution* if

$$u(x+z,t) - u(x,t) \;\le\; \frac{z}{t} \qquad \text{for a.e. } (x,t) \in \mathbb{R} \times (0,\infty) \text{ and all } z > 0. \tag{24}$$

Condition (24) is Oleinik's one-sided Lipschitz bound [12].

Lemma 6.3 (Oleinik implies Lax)**.** *Let u be a piecewise C^1 entropy solution. Then every jump of u satisfies the Lax condition* (22)*; for Burgers' equation, $u_\ell > u_r$ at every jump.*

Proof. Fix a curve $x = s(t)$ with $u_\ell(t) \neq u_r(t)$. Since u is continuous on each side of the curve, (24) may be applied (for a.e. t) with $x \uparrow s(t)$ and $x + z \downarrow s(t)$, so $z \to 0$. The left side tends to $u_r - u_\ell$ and the right side to 0; hence $u_r - u_\ell \le 0$, and since the limits differ, $u_\ell > u_r$. By continuity of u_ℓ, u_r in t this holds for every $t > 0$. □

Remark 6.4. For a piecewise C^1 solution with locally finitely many jumps, (24) holds if and only if each jump satisfies $u_\ell > u_r$ and $u_x \le 1/t$ on the smooth pieces. The Lax condition alone therefore does not characterise entropy solutions among arbitrary L^∞ weak solutions.

Proposition 6.5. *For the Riemann data* (19) *($u_\ell = 0 < u_r = 1$), the rarefaction solution u_B of* (21) *is an entropy solution, while the shock solution u_A of* (20) *and every u^α of Remark 5.4 are not.*

Proof. *u_B is entropy.* Write $u_B(x,t) = \min\{\max\{x/t, 0\}, 1\}$, the composition of $x \mapsto x/t$ with the 1-Lipschitz function $y \mapsto \min\{\max\{y, 0\}, 1\}$. Hence $u_B(\cdot, t)$ is $(1/t)$-Lipschitz and $u_B(x+z,t) - u_B(x,t) \le z/t$, which is (24).

u_A is not. The jump $0 \to 1$ has $u_\ell < u_r$, violating (23), so by Lemma 6.3 u_A cannot be an entropy solution. Directly: for $x = t/2 - z/2$, $u_A(x+z,t) - u_A(x,t) = 1 > z/t$ as soon as $z < t$.

u^α is not. Each u^α contains an upward jump: if $\alpha > 0$ the jump $0 \to \alpha$ is upward, and if $\alpha \le 0$ the jump $\alpha \to 1$ is. The same argument as for u_A applies. □

6.3 Uniqueness

Theorem 6.6 (Oleinik; [5, §3.4.3])**.** *For $u_0 \in L^\infty(\mathbb{R})$ and $f(u) = u^2/2$, the initial value problem $u_t + f(u)_x = 0$, $u(\cdot, 0) = u_0$, has at most one entropy solution in the sense of Definition 6.2, up to a set of measure zero. An entropy solution exists: the Lax–Oleinik solution [5, §3.4.2] is a weak solution satisfying* (24) *(Remark 8.5).*

For a general uniformly convex flux, Oleinik's bound reads $u(x+z,t) - u(x,t) \le C(1 + 1/t)\, z$ for some constant C, and uniqueness is proved in [5, §3.4.3] in this larger class. Since $z/t \le C(1 + 1/t)z$ with $C = 1$, every solution satisfying (24) lies in that class, so uniqueness there implies the statement above. For Burgers' equation the two classes coincide: any solution in the larger class equals the Lax–Oleinik solution, which satisfies (24) (Remark 8.5). The proof proceeds by duality [5]; Kruzhkov's formulation via entropy inequalities [8] yields an L^1-contraction and covers non-convex fluxes. It is not reproduced here.

7 The Riemann Problem

The entropy solution is now determined for arbitrary constant states $u_\ell, u_r \in \mathbb{R}$, that is, for the data

$$u_0(x) = \begin{cases} u_\ell, & x < 0, \\ u_r, & x > 0. \end{cases} \tag{25}$$

7.1 The rarefaction fan

Seek self-similar solutions $u(x,t) = g(\xi)$, $\xi = x/t$. Then

$$u_t + u\,u_x = -\frac{\xi}{t}g'(\xi) + \frac{g(\xi)}{t}g'(\xi) = \frac{g'(\xi)}{t}\big(g(\xi) - \xi\big) = 0, \tag{26}$$

so on each interval of ξ either $g'(\xi) = 0$ (a constant state) or $g(\xi) = \xi$ (the fan $u = x/t$). The fan connects the states $u_\ell < u_r$.

7.2 The complete solution

Theorem 7.1 (Entropy solution of the Riemann problem)**.** *The unique entropy solution of* (3) *with data* (25) *is:*

(a) *If $u_\ell > u_r$ (shock case),*

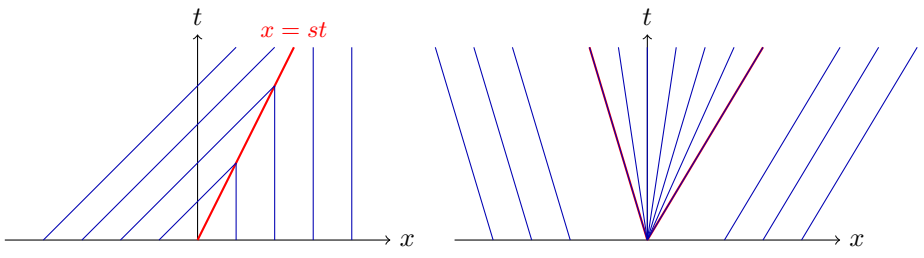
$$u(x,t) = \begin{cases} u_\ell, & x < st, \\ u_r, & x > st, \end{cases} \qquad s = \frac{u_\ell + u_r}{2}. \tag{27}$$

(b) *If $u_\ell \le u_r$ (rarefaction case),*

$$u(x,t) = \begin{cases} u_\ell, & x \le u_\ell t, \\ x/t, & u_\ell t < x < u_r t, \\ u_r, & x \ge u_r t. \end{cases} \tag{28}$$

Proof. *(a)* Apply Proposition 4.5 with $k = 1$, $s_1(t) = st$: u is constant on each side (so (i) holds), the jump moves at the Rankine–Hugoniot speed $s = (u_\ell + u_r)/2$ (so (ii) holds), and for $x < 0$ (resp. $x > 0$) we have $x < st$ (resp. $x > st$) for small t, giving (iii). So u is a weak solution. It is an entropy solution: the only possible increments of $u(\cdot, t)$ are 0 and $u_r - u_\ell < 0$, so $u(x+z,t) - u(x,t) \le 0 \le z/t$, so (24) holds. Uniqueness is Theorem 6.6. The characteristics are shown in Figure 2(a).

(b) The case $u_\ell = u_r$ is the trivial constant solution, so suppose $u_\ell < u_r$. Apply Proposition 4.5 with $k = 2$, $s_1(t) = u_\ell t$, $s_2(t) = u_r t$ (starting at a common point). (i) u is constant on the outer regions and equals x/t on the fan $u_\ell t < x < u_r t$, where it solves the PDE by Section 7.1. (ii) u is continuous across both curves, so (17) holds trivially. (iii) For fixed $x < 0$ we have $x < u_\ell t$ for all small t, and for fixed $x > 0$, $x > u_r t$ for all small t, so $u(\cdot, t) \to u_0$ away from 0. Hence u is a weak solution. It is an entropy solution because $u(x,t) = \min\{\max\{x/t, u_\ell\}, u_r\}$ is $(1/t)$-Lipschitz in x, so (24) holds. Uniqueness is again Theorem 6.6; see Figure 2(b). □



(a) shock: $u_\ell > u_r$ (b) rarefaction: $u_\ell < u_r$

Figure 2: Characteristics for the two cases of Theorem 7.1. (a) $u_\ell = 1$, $u_r = 0$, $s = 1/2$: characteristics on both sides run into the shock $x = st$. (b) Characteristics fan out from the origin.

8 Extensions

8.1 The Cole–Hopf transformation

Theorem 8.1 (Cole–Hopf transformation [3, 7])**.** *Let u be a classical solution of $u_t + u\,u_x = \nu\,u_{xx}$ on $\mathbb{R}\times[0,T)$, and set*

$$\hat\psi(x,t) = -\frac{1}{2\nu}\int_0^x u(y,t)\,dy, \qquad c(t) = \int_0^t \Big(\frac{u(0,\tau)^2}{4\nu} - \frac{u_x(0,\tau)}{2}\Big)d\tau, \qquad \theta(x,t) = \exp\big(\hat\psi(x,t)+c(t)\big) > 0. \tag{29}$$

Then θ solves the heat equation $\theta_t = \nu\,\theta_{xx}$, and $u = -2\nu\,\theta_x/\theta$.

Proof. Since c depends on t only, $\theta_x/\theta = \hat\psi_x = -u/(2\nu)$, which gives $u = -2\nu\theta_x/\theta$. For the heat equation, let $w = \hat\psi_t - \nu\hat\psi_{xx} - \nu\hat\psi_x^2$. Using $\hat\psi_x = -u/(2\nu)$, $\hat\psi_{xx} = -u_x/(2\nu)$, $\hat\psi_{xt} = -u_t/(2\nu)$ and $\hat\psi_{xxx} = -u_{xx}/(2\nu)$,

$$w_x = \hat\psi_{xt} - \nu\hat\psi_{xxx} - 2\nu\hat\psi_x\hat\psi_{xx} = -\frac{1}{2\nu}\big(u_t + u\,u_x - \nu\,u_{xx}\big) = 0. \tag{30}$$

Hence w depends on t only and may be evaluated at $x=0$, where $\hat\psi(0,t)=0$ gives $\hat\psi_t(0,t)=0$:

$$w(t) = w(0,t) = \frac{u_x(0,t)}{2} - \frac{u(0,t)^2}{4\nu} =: C(t). \tag{31}$$

With $\psi = \hat\psi + c$ we have $c' = -C$, so $\psi_t - \nu\psi_{xx} - \nu\psi_x^2 = C + c' = 0$. Since $\theta = e^{\psi}$ satisfies $\theta_t = \theta\psi_t$ and $\theta_{xx} = \theta(\psi_{xx}+\psi_x^2)$, this is $\theta_t = \nu\theta_{xx}$. □

Remark 8.2. The factor $e^{c(t)}$ is independent of x, so it affects neither $u = -2\nu\theta_x/\theta$ nor the initial data ($c(0)=0$). It cannot be omitted: $e^{\hat\psi}$ satisfies $\partial_t\theta = \nu\theta_{xx} + C(t)\theta$, which is the heat equation only if $C\equiv 0$.

8.2 Exact solution via the heat kernel

By Theorem 8.1, (1) with bounded data u_0 reduces to the heat equation for θ with initial data

$$\theta_0(x) = \exp\left(-\frac{1}{2\nu}\int_0^x u_0(y)\,dy\right) \le e^{\|u_0\|_\infty |x|/(2\nu)}. \tag{32}$$

The Cauchy problem for the heat equation has a unique solution in this growth class [5, §2.3.3], namely

$$\theta(x,t) = \int_{\mathbb{R}} \frac{1}{\sqrt{4\pi\nu t}}\,\exp\left(-\frac{(x-y)^2}{4\nu t}\right)\theta_0(y)\,dy. \tag{33}$$

Substituting (32)–(33) into $u = -2\nu\,\theta_x/\theta$ and differentiating under the integral gives the *Cole–Hopf formula*

$$u(x,t) = \frac{\displaystyle\int_{\mathbb{R}} \frac{x-y}{t}\,\exp\left(-\frac{G(x,y,t)}{2\nu}\right)dy}{\displaystyle\int_{\mathbb{R}} \exp\left(-\frac{G(x,y,t)}{2\nu}\right)dy}, \qquad G(x,y,t) = \int_0^y u_0(s)\,ds + \frac{(x-y)^2}{2t}. \tag{34}$$

Corollary 8.3 (Global smoothness; Hopf [7])**.** *For every bounded measurable u_0 and every $\nu>0$, formula* (34) *defines a solution of* (1) *that is C^∞ for all $t>0$, and it is the unique bounded solution with data u_0.*

That (34) solves the equation follows by reversing the computation of Theorem 8.1: if $\theta>0$ solves the heat equation, then $u = -2\nu\theta_x/\theta$ solves (1), and θ in (33) is smooth and positive for $t>0$. Attainment of the initial data and uniqueness of the bounded solution for merely measurable u_0 are due to Hopf [7] and are cited without proof.

8.3 The vanishing viscosity limit

Theorem 8.4 (Vanishing viscosity limit [7])**.** *Let $u_0 \in L^\infty(\mathbb{R})$, let u^ν be the solution of* (1) *with data u_0 given by* (34)*, and let u be the entropy solution of* (2) *with the same data (Definition 6.2; for Riemann data, Theorem 7.1). Then $u^\nu \to u$ pointwise almost everywhere and in L^1_{loc} as $\nu\to 0^+$.*

The proof is a Laplace-method analysis of (34) and is not reproduced. The mechanism is as follows.

Remark 8.5 (The Lax–Oleinik formula). As $\nu \to 0$ the weight $\exp(-G/2\nu)$ in (34) concentrates at the minimisers of $y \mapsto G(x,y,t)$. Where the minimiser $y^*(x,t)$ is unique, the limit is

$$u(x,t) = \frac{x - y^*(x,t)}{t}, \qquad y^*(x,t) = \arg\min_y \Big\{ V(y) + \frac{(x-y)^2}{2t} \Big\}, \quad V(y) = \int_0^y u_0, \tag{35}$$

the *Lax–Oleinik formula* [9, 12], [5, §3.4.2]; a jump occurs where two minimisers tie. Since $x \mapsto y^*(x,t)$ is nondecreasing [5, §3.4.2], for $z > 0$

$$u(x+z,t) - u(x,t) = \frac{z - \big(y^*(x+z,t) - y^*(x,t)\big)}{t} \;\le\; \frac{z}{t},$$

which is (24). As (35) also defines a weak solution [5, §3.4.2], the Lax–Oleinik solution is an entropy solution in the sense of Definition 6.2; by Lemma 6.3 every jump is downward.

8.3.1 Viscous shock profiles

Proposition 8.6 (Viscous shock profiles)**.** *Let $u_\ell \neq u_r$ and $\nu > 0$. Equation* (1) *has a travelling-wave solution $u(x,t) = \phi(x - st)$, with $\phi \in C^2$ bounded, $\phi(\xi) \to u_\ell$ as $\xi \to -\infty$, $\phi(\xi) \to u_r$ as $\xi \to +\infty$ and $\phi' \to 0$, if and only if $u_\ell > u_r$. In that case $s = (u_\ell + u_r)/2$ and, with $\Delta = u_\ell - u_r$,*

$$\phi(\xi) = s - \frac{\Delta}{2} \tanh\Big(\frac{\Delta\,\xi}{4\nu}\Big). \tag{36}$$

Moreover $\|\phi - (s - \frac{\Delta}{2}\operatorname{sgn}\xi)\|_{L^1(\mathbb{R})} = 4\ln 2\;\nu$, independent of s and Δ.

Proof. Substituting $u = \phi(x - st)$ gives $-s\phi' + \phi\phi' = \nu\phi''$, and integrating once from $-\infty$ (where $\phi' \to 0$),

$$\nu\phi' = \tfrac{1}{2}\phi^2 - s\phi - C, \qquad C = \tfrac{1}{2}u_\ell^2 - s u_\ell.$$

Letting $\xi \to +\infty$ shows that the right-hand side must also vanish at u_r, so u_ℓ, u_r are the two roots of the quadratic: $s = (u_\ell + u_r)/2$ (the Rankine–Hugoniot speed) and

$$\nu\phi' = \tfrac{1}{2}(\phi - u_\ell)(\phi - u_r). \tag{37}$$

This autonomous first-order ODE cannot be crossed at its equilibria u_ℓ, u_r, so a connecting ϕ lies strictly between them. On $(\min(u_\ell,u_r), \max(u_\ell,u_r))$ the right side of (37) is negative, so ϕ is decreasing. If $u_\ell < u_r$ it would have to increase from u_ℓ to u_r, a contradiction; if $u_\ell > u_r$, (37) is solved by (36). For the L^1 distance, by oddness and $\int_0^\infty (1 - \tanh\sigma)\,d\sigma = \ln 2$,

$$\int_{\mathbb{R}} \tfrac{\Delta}{2}\Big|\tanh\tfrac{\Delta\xi}{4\nu} - \operatorname{sgn}\xi\Big|\,d\xi = \Delta\int_0^\infty \Big(1 - \tanh\tfrac{\Delta\xi}{4\nu}\Big)d\xi = \Delta\cdot\frac{4\nu}{\Delta}\ln 2 = 4\ln 2\,\nu.$$

□

Hence a jump is the limit of viscous travelling waves, of thickness $O(\nu/\Delta)$, exactly when $u_\ell > u_r$. This recovers the Lax condition (23) from vanishing viscosity, and shows that the upward jump of u_A in Section 5 is not a vanishing-viscosity limit.

8.4 Godunov's method

8.4.1 The scheme

Let $\mathbb{R}$ be divided into cells $[x_{j-1/2}, x_{j+1/2}]$ of width Δx, with $t^n = n\Delta t$ and cell averages $U_j^n \approx \Delta x^{-1}\int_{x_{j-1/2}}^{x_{j+1/2}} u(x,t^n)\,dx$. Integrating $u_t + f(u)_x = 0$ over a cell and a time step gives

$$U_j^{n+1} = U_j^n - \frac{\Delta t}{\Delta x}\left(\widehat{f}_{j+1/2} - \widehat{f}_{j-1/2}\right), \tag{38}$$

with $\widehat{f}_{j+1/2}$ the time-averaged flux through $x_{j+1/2}$. Godunov's method [6] takes $\widehat{f}_{j+1/2}$ to be the flux at $x/t = 0$ of the exact Riemann solution with data (U_j^n, U_{j+1}^n) (Theorem 7.1):

$$\widehat{f}_{j+1/2} = f\big(u^R(0;\,U_j^n, U_{j+1}^n)\big), \tag{39}$$

where $u^R(\xi; u_\ell, u_r)$ is the entropy solution with data (u_ℓ, u_r) at $\xi = x/t$. By (27)–(28),

$$u^R(0; u_\ell, u_r) = \begin{cases} u_\ell, & u_\ell > u_r,\ \frac{u_\ell+u_r}{2} > 0, \\ u_r, & u_\ell > u_r,\ \frac{u_\ell+u_r}{2} < 0, \\ u_\ell, & u_\ell \le u_r,\ u_\ell \ge 0, \\ u_r, & u_\ell \le u_r,\ u_r \le 0, \\ 0, & u_\ell < 0 < u_r, \end{cases} \tag{40}$$

the last case corresponding to $\xi = 0$ lying inside the fan. If $u_\ell = -u_r$ the value $u^R(0)$ is ambiguous, but $f(u_\ell) = f(u_r)$, so (39) is unaffected.

8.4.2 Stability and convergence

The scheme is stable and monotone under the CFL condition

$$\Delta t \le \frac{\Delta x}{\max_j |U_j^n|}, \tag{41}$$

which ensures that waves from neighbouring interfaces do not reach $x_{j+1/2}$ within one step [10, Ch. 4].

Theorem 8.7 ([4]; [10, Ch. 12])**.** *Under* (41)*, the Godunov approximations converge, as* $\Delta x, \Delta t \to 0$*, to the unique entropy solution of Theorem 6.6.*

The theorem asserts convergence, not a rate.

8.4.3 Numerical results

The scheme was implemented (`code/godunov.py`) on $x \in [-2, 2]$ with N cells, cell averages of the Riemann data (the discontinuity lies on a cell interface), outflow boundary conditions and $\Delta t = 0.45\,\Delta x / \max_j |U_j^n|$, up to $t = 1$, for the rarefaction data $(u_\ell, u_r) = (0, 1)$ and the shock data $(1, 0)$. Errors are measured in $L^1(-2, 2)$ against the exact solution of Theorem 7.1, integrated cell by cell. The case-based flux (39) agrees with the closed form $\min_{[u_\ell, u_r]} f$ (if $u_\ell \le u_r$) and $\max_{[u_r, u_\ell]} f$ (otherwise) on 10^5 random state pairs, and total mass equals the initial mass plus the net boundary flux to ten digits. Figure 3 shows $N = 400$: the computed shock lies at $x = st = 0.5$ and the fan has slope $1/t$, with no oscillation.

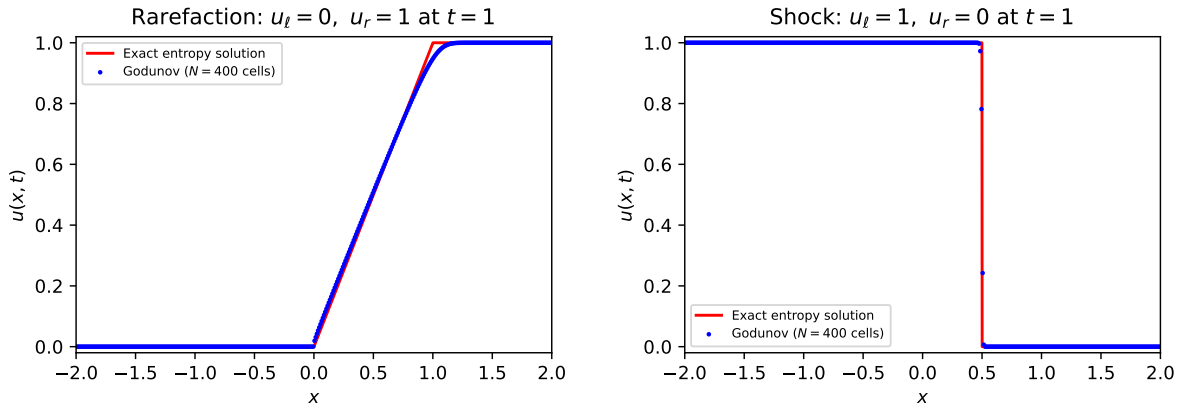


Figure 3: Godunov scheme (dots, $N = 400$) against the exact entropy solution (solid) at $t = 1$. Left: rarefaction data $(0, 1)$. Right: shock data $(1, 0)$.

Shock. The shock $x = st = 0.5$ lies on a cell interface iff $8 \mid N$. For such N the error is $0.50\,\Delta x$ to two digits (0.495–0.501) and the observed order is ≈ 1 (Table 1). For $N = 97, \dots, 103$ the error ranges from $0.60\,\Delta x$ to $0.77\,\Delta x$ (e.g. $0.655\,\Delta x$ at $N = 100$, against $0.50\,\Delta x$ at $N = 96, 104$), so orders computed between misaligned and aligned resolutions are unreliable ($N = 100 \to 200$ would give 1.40); the table therefore uses $N = 200 \cdot 2^k$. In all cases the error is proportional to Δx.

N	Δx	L^1 (rarefaction)	order	L^1 (shock)	order
200	2.0×10^{-2}	3.05×10^{-2}	—	9.91×10^{-3}	—
400	1.0×10^{-2}	1.82×10^{-2}	0.74	4.98×10^{-3}	0.99
800	5.0×10^{-3}	1.07×10^{-2}	0.77	2.50×10^{-3}	0.99
1600	2.5×10^{-3}	6.14×10^{-3}	0.80	1.24×10^{-3}	1.01
3200	1.25×10^{-3}	3.48×10^{-3}	0.82	6.23×10^{-4}	0.99
6400	6.25×10^{-4}	1.95×10^{-3}	0.84	3.13×10^{-4}	0.99

Table 1: L^1 errors at $t=1$ and observed orders $\log_2(e_{N/2}/e_N)$. All N are multiples of 8, so the shock lies on a cell interface.

Rarefaction. The error decreases monotonically with observed order $0.74, 0.77, 0.80, 0.82, 0.84$, still increasing. A first-order rate is therefore not established by these experiments. The excess error is attributable to the two corners of the fan, where the exact solution is only Lipschitz.

8.4.4 Numerical diffusion as vanishing viscosity

For the shock data all states lie in $[0,1]$ and $s=\frac12>0$, so the Godunov flux is the upwind flux and the scheme reads $U_j^{n+1}=U_j^n-\lambda(f(U_j^n)-f(U_{j-1}^n))$, $\lambda=\Delta t/\Delta x$. Taylor expansion gives the modified equation

$$u_t+u\,u_x=\frac{\Delta x}{2}\big(u(1-\lambda u)\,u_x\big)_x+O(\Delta x^2), \tag{42}$$

a viscous Burgers' equation with viscosity $\nu(u)=\frac{\Delta x}{2}u(1-\lambda u)$, nonnegative exactly under (41). Its travelling shock $1\to0$ of speed $\frac12$ satisfies $\nu(u)u_\xi=\frac12u(u-1)$ in $\xi=x-t/2$; in units of Δx,

$$\frac{du}{d\xi}=\frac{u-1}{1-\lambda u},\qquad \xi(u)=\xi_0+(1-\lambda)\ln(1-u)-\lambda u. \tag{43}$$

The profile decays exponentially to 1 as $\xi\to-\infty$ and reaches 0 at a finite ξ_0, which conservation of mass fixes as $\xi_0=1-\lambda/2$. The L^1 distance from the sharp shock is then

$$\frac{\|u-u_{\text{exact}}\|_{L^1}}{\Delta x}=2\int_0^{u_0}\frac{u(1-\lambda u)}{1-u}\,du,\qquad (1-\lambda)\ln(1-u_0)-\lambda u_0=-\Big(1-\frac{\lambda}{2}\Big). \tag{44}$$

Table 2 compares this with the measured shock error on aligned grids, where λ equals the CFL number since $\max|U|=1$ (`code/numerical_viscosity.py`).

CFL $=\lambda$	measured ($N=1600$)	predicted	difference
0.10	0.627	0.686	+9.4%
0.20	0.589	0.635	+8.0%
0.30	0.553	0.586	+5.8%
0.45	0.496	0.511	+3.1%
0.60	0.448	0.438	−2.4%
0.80	0.352	0.342	−3.0%
0.95	0.300	0.272	−9.3%

Table 2: Shock L^1 error divided by Δx: measured (grids with $8\mid N$) against the modified-equation prediction.

The prediction is within 3% at CFL 0.45 and within 10% throughout. The value at CFL 0.95 is not yet asymptotic (0.334 at $N=400$, 0.300 at $N=1600$). Equating the error with the viscous-shock value $4\ln2\,\nu_{\text{eff}}$ of Proposition 8.6 ($s=\frac12$, $\Delta=1$) defines an effective viscosity: the measured $0.50\,\Delta x$ gives $\nu_{\text{eff}}\approx0.180\,\Delta x$, the modified-equation profile gives $0.184\,\Delta x$, and $\nu(\frac12)=\frac{\Delta x}{2}\cdot\frac12(1-\frac{\lambda}{2})$ gives $0.194\,\Delta x$ at $\lambda=0.45$. Since $\nu_{\text{eff}}=O(\Delta x)\to0$, the scheme acts in the shock layer as a vanishing viscosity, in accordance with Theorem 8.4. This is a consistency check rather than a proof: the numerical shock spans a few cells and (42) omits the $O(\Delta x^2)$ terms.

9 Conclusion

The weak formulation extends solutions of (2) beyond breakdown but does not determine them: already for Riemann data there are infinitely many weak solutions (Section 5). The Oleinik condition, whose local form at a jump is the

Lax condition $u_\ell > u_r$, restores uniqueness (Sections 6–7). It is the physically correct selection because it is the one made by vanishing viscosity: the Cole–Hopf solutions converge to the entropy solution (Theorem 8.4, Remark 8.5), and a viscous travelling wave connecting u_ℓ to u_r exists if and only if $u_\ell > u_r$ (Proposition 8.6). The underlying theory is standard [5, §3.4].

The Godunov scheme of Section 8.4 was tested on Riemann data. The shock error is $0.50\,\Delta x$ on grids aligned with the shock and 0.60–$0.77\,\Delta x$ otherwise; the rarefaction error has observed order ≈ 0.8, still increasing. The modified equation (42) predicts the shock error to within 3% at CFL 0.45 and within about 10% for CFL numbers in $[0.1, 0.95]$, corresponding to an effective viscosity $\nu_{\text{eff}} \approx 0.18\,\Delta x$. This is a heuristic consistency check; the rigorous result is Theorem 8.7 [4].

The proofs of Theorems 6.6, 8.4 and 8.7 are cited, not reproduced, and the long-time N-wave asymptotics for compactly supported data [9], [5, §3.4.5] are not treated.

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